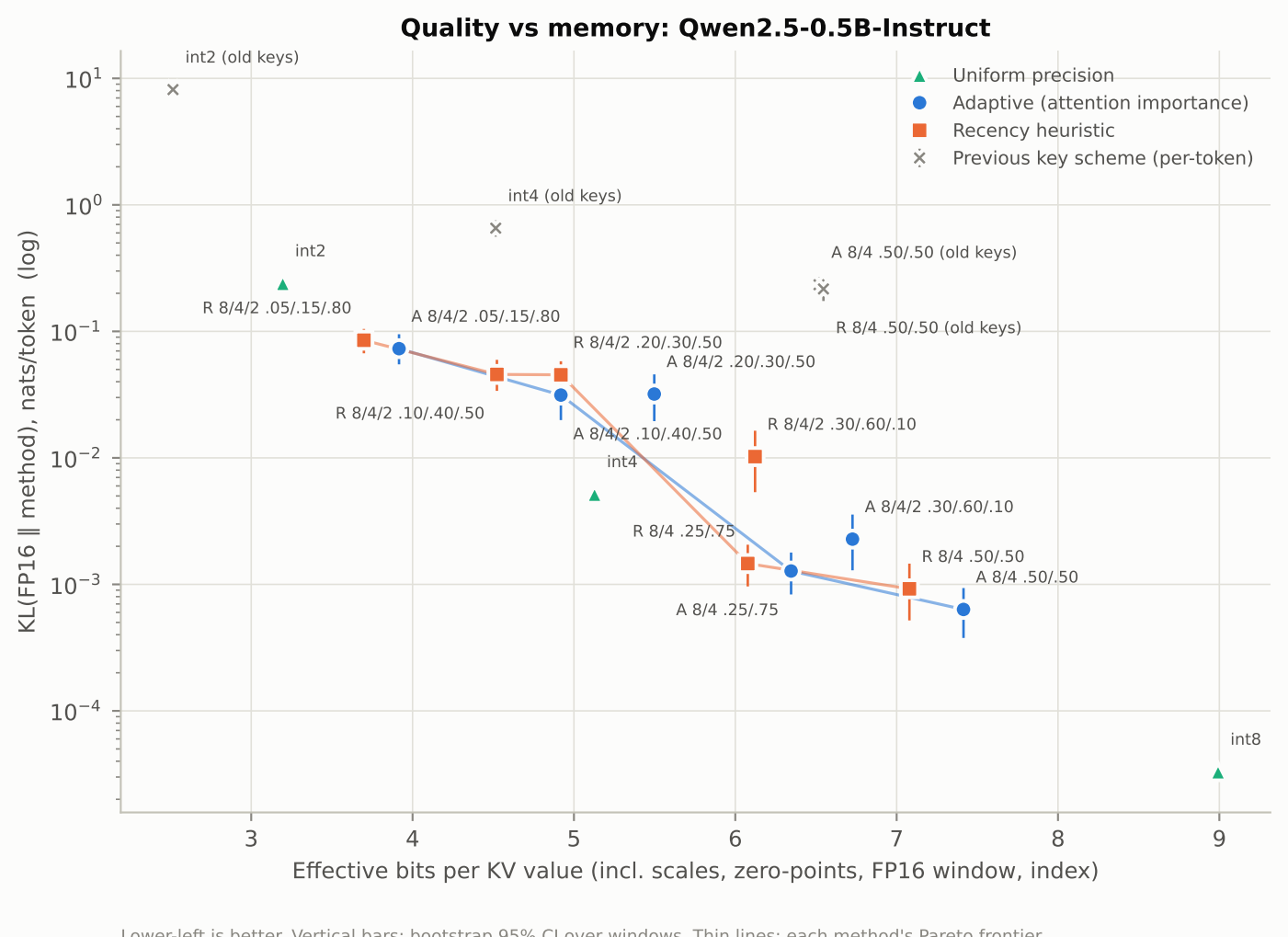
Quality vs memory: Qwen2.5-0.5B-Instruct
Uniform precision
Adaptive (attention importance)
Recency heuristic
Previous key scheme (per-token)
int2 (old keys)
int4 (old keys)
int2
A 8/4 .50/.50 (old keys)
R 8/4/2 .05/.15/.80
A 8/4/2 .05/.15/.80
R 8/4 .50/.50 (old keys)
R 8/4/2 .20/.30/.50
A 8/4/2 .20/.30/.50
R 8/4/2 .10/.40/.50
A 8/4/2 .10/.40/.50
R 8/4/2 .30/.60/.10
int4
A 8/4/2 .30/.60/.10
R 8/4 .25/.75
R 8/4 .50/.50
A 8/4 .50/.50
A 8/4 .25/.75
int8
KL(FP16 ‖ method), nats/token (log)
Effective bits per KV value (incl. scales, zero-points, FP16 window, index)
Lower-left is better. Vertical bars: bootstrap 95% CI over windows. Thin lines: each method's Pareto frontier.